Inorganic Cume

Sept 12, 1998

70%

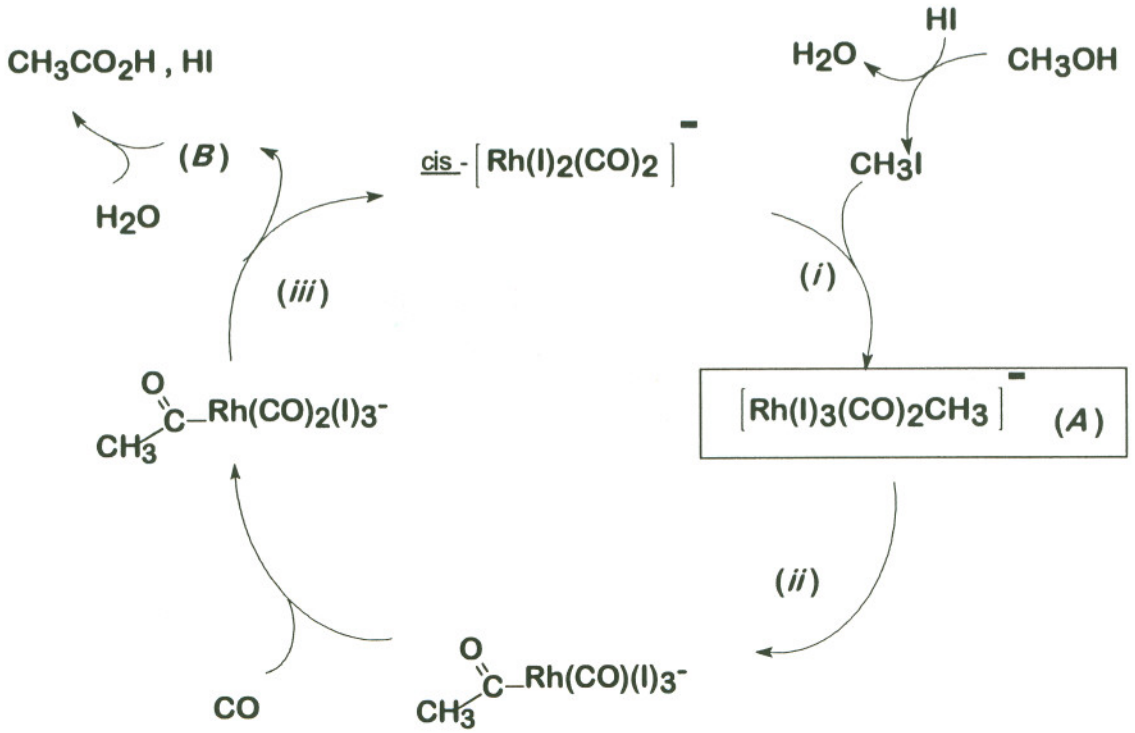
1) For the catalytic cycle below, give the appropriate structural formulas for the intermediate (A) and first product (B). Give the general category (*e.g. acid-base, substitution,..)* and balanced reactions for the steps labeled (i) , (ii) , and (iii). Give the electron counts for the three complexes

$[Rh(CO)_2(I)_2]^{1-}$ , (*at the top of the cycle*)

(*A*) ,

$[Rh(CH_3CO)(CO)(I)_3]^{1-}$ .. (*at the bottom of the cycle*)

What is the overall reaction that is catalyzed by this rhodium complex? If acetic acid was added in the conversion of (B) instead of H2O, what product would be formed?

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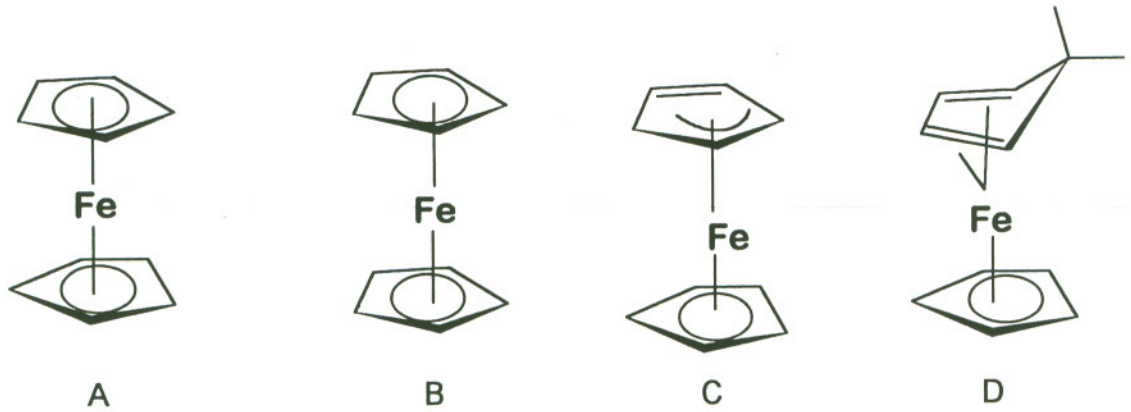
2) Distinguish between the four related but different complexes below. I.e. , what differences in structure or properties are implied by these drawings. Give the electron counts for A - D.